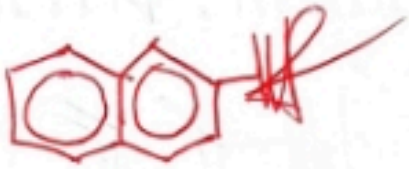
PROPOSED SCORING GUIDE

JUNE, 2026

CHEMISTRY THEORY [525/1]

BY

CHRISTOPHER



0746332976 → Whatsapp

0725967573 → Calls

MWANGA JOHN CHE...



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ITEM 1

$Zn_{(s)} + 2e \longrightarrow Zn^{2+}_{(aq)}$ $+0.76$.

$Cu_{(s)} + 2e \longrightarrow Cu^{2+}_{(aq)}$ -0.34

All are undergoing Oxidation

Make them to undergo Reduction.

$Zn^{2+}_{(aq)} \longrightarrow Zn_{(s)} + 2e$ -0.76.

$Cu^{2+}_{(aq)} \longrightarrow Cu_{(s)} + 2e$ $+0.34$

Therefore, Zinc is the more negative reduction Potential and therefore, it loses electrons more readily hence it is the stronger reducing agent than Copper.

Copper has a more positive reduction Potential and therefore, it gains electrons more readily hence it is the stronger oxidizing agent than Zinc.

$E^{\ominus}_{cell} = E^{\ominus}_{cathode} - E^{\ominus}_{anode}$.

$= +0.34V - -0.76$

$E^{\ominus}cell. = +1.10V$: $E^{\ominus}cell$ is greater than zero. The cell reaction is spontaneous or pasable.

Equation for overall. ; Accept without

$Cu^{2+}_{(aq)} + Zn_{(s)} \longrightarrow Cu_{(s)} + Zn^{2+}_{(aq)}$

Cell convention.

$Zn_{(s)} | Zn^{2+}_{(aq)} || Cu^{2+}_{(aq)} | Cu_{(s)}$

For Zinc, Zn.

Total ionisation energy $= 1^{st} + 2^{nd}$.

$= (906 + 1733.3)\ KJmol^{-1}$

$= +2639.3\ KJmol^{-1}$.

Da

Hydration energy $= -2046\ KJmol^{-1}$

Net energy $= (2639.3 + -2046)$

$= +593.3\ KJmol^{-1}$

Da.

For copper.

Total Ionisation $= 1^{st} + 2^{nd}$.

$= 754 + 1958$

$= +2712\ KJmol^{-1}$

Da.

Hydration energy $= +2100\ KJmol^{-1}$

Net energy $= (2712 + +2100)$

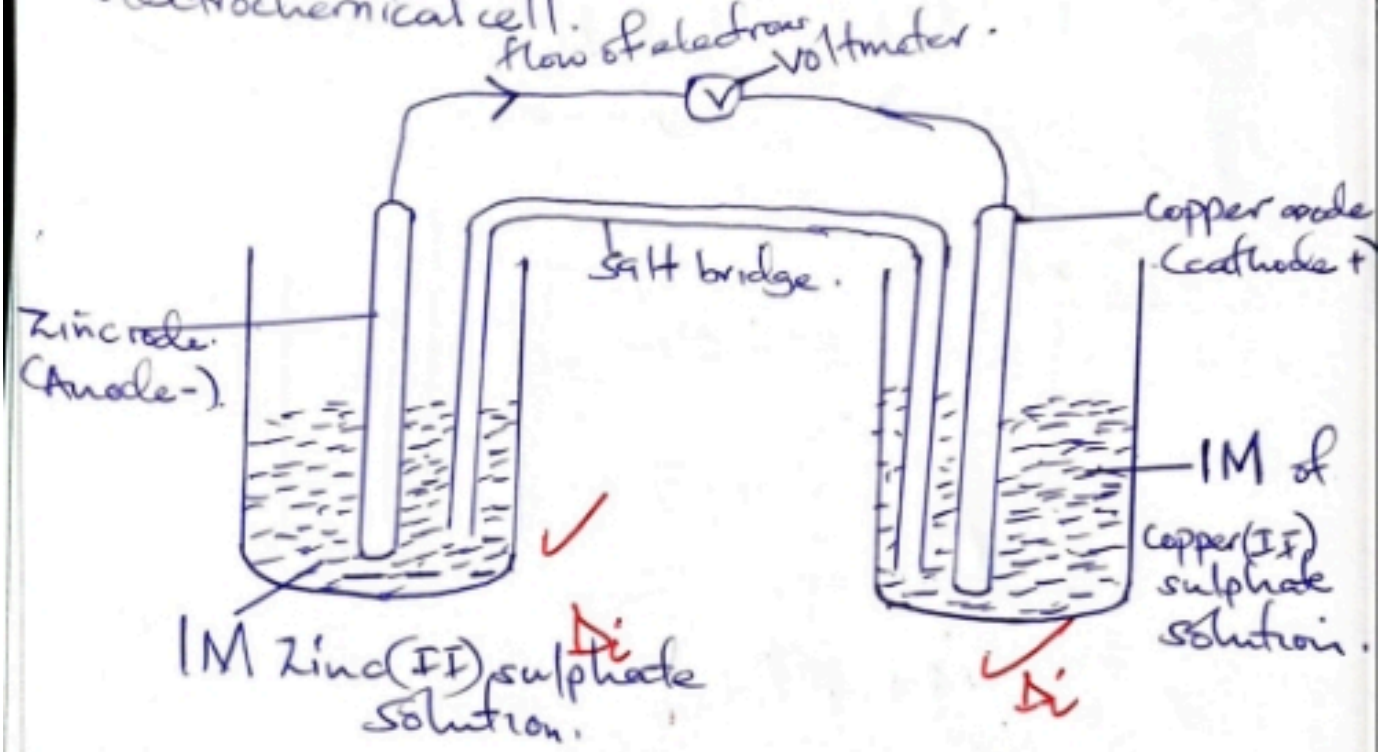
Da.

$= +612.0\ KJmol^{-1}$

Therefore, Copper requires more energy to form hydrated ions, aligning with it's less reactive nature and positive standard reduction potential

Compared to Zinc.

Electrochemical cell.



TOTAL SCORES: 15

5 SCORES TRANSFERED TO SECTION B.

ACCEPT 20 SCORES

ITEM 2

Percentage of nitrogen, N

$= 100 - (77.3 + 7.58)$

$= 15.12\%$ ✓ Da.

Element	C	H	N.
Percentage composition	77.3	7.58	15.12
moles	$\frac{77.3}{12}$ 6.4417	$\frac{7.58}{1}$ 7.58	$\frac{15.12}{14}$ 1.08
Mole ratio	$\frac{6.4417}{1.08}$ 6.	$\frac{7.58}{1.08}$ 7	$\frac{1.08}{1.08}$ 1

Emprical formula. $C_6H_7N_1$ Da.

Molecular mass of water, H_2O $= (2 \times 1) + (1 \times 16)$

$= 18_g$.

$P_{total} = P_{water} + P_x$ ✓

$760 = 734.9 + P_x$. ✓

$P_x = 25.1 mm Hg$. Da.

Composition of water = 85.
Composition of X = 15.

$$\frac{\text{mass of X}}{\text{mass of water}} = \frac{P_x \times M_{r_x}}{P_{H_2O} \times M_{r_{H_2O}}}$$ ✓ ✓

$$\frac{15}{85} = \frac{25.1 \times M_{r_x}}{734.9 \times 18}$$

$$0.1765 = \frac{25.1 M_{r_x}}{13228.2}$$ ✓ Di

$$M_{r_x} = \frac{0.1765 \times 13228.2}{25.1}$$

$$Mr_x = 93_g$$ ✓ Da

Molecular mass of X_1 = 93g.

$$(C_6H_7N)_n = 93$$

$$93n = 93$$

$$n = \frac{93}{93}$$ ✓ Da.

$$= 1$$

Molecular formula is C_6H_7N ✓ Dc

Applications of steam distillation.

Extraction of natural products from plant materials such as essential oils used in perfumes, fragrances and personal hygience products

Purifying temperature, sensitive organic compounds below their natural boiling points to avoid decomposition

TOTAL SCORES 15

5 SCORES TRANSFERED TO SECTION B

ACCEPT 20 SCORES

ITEM 3.

Periodic trends affect reactivity and reaction of elements with water, alkali and dilute acids, health implications of any of the elements.

Trend on the Ionisation energy

Generally Ionisation energy increases from sodium to chlorine.

From sodium to chlorine, there is an increase in nuclear charge because a proton is added to the nucleus from element to element. Screening effect almost remains constant because an electron is added to the same energy level.

The increase in nuclear charge, increases effective nuclear charge thus decrease in atomic radius and as a result, the outer most electrons are strongly attracted than the repelled by the nucleus. Thus increasing the Ionisation energy of the element.

Trend on electron affinity

Generally, electron affinity increases from sodium to chlorine.

From sodium to chlorine, nuclear charge increases because a proton is added to the nucleus from element to element. Screening effect almost remains constant because electrons are being added to the same energy level. The increase in nuclear charge increases on effective nuclear charge thus decrease in the atomic radius of the element.

Therefore, this increase on the attraction of the incoming electron hence increase in electron affinity

Reaction with water.

Sodium being a reducing agent, it reacts vigorously with cold water to form sodium hydroxide and hydrogen gas in a redox reaction. ✓re

$$Na_{(s)} + H_2O_{(l)} \longrightarrow NaOH_{(aq)} + \tfrac{1}{2}H_{2(g)}$$ ✓eq

Magnesium being a reducing agent, it reacts slowly with cold water in a redox reaction forming magnesium hydroxide and hydrogen gas ✓re

$$Mg_{(s)} + 2H_2O_{(l)} \longrightarrow Mg(OH)_{2(aq)} + H_{2(g)}.$$ ✓eq

It also reacts with steam on heating to form magnesium oxide and hydrogen gas in a redox reaction. ✓re

$$Mg_{(s)} + H_2O_{(g)} \longrightarrow MgO_{(s)} + \tfrac{1}{2}H_{2(g)}$$ ✓eq

Aluminium does not react with water because it forms a protective layer of Aluminium oxide.

However, Aluminium being highly reducing it reacts with steam on heating in a redox reaction to form Aluminium oxide and hydrogen gas. ✓re

$$2Al_{(s)} + 3H_2O_{(g)} \longrightarrow Al_2O_{3(s)} + 3H_{2(g)}$$ ✓eq

Silicon, being a reducing agent, it reacts with heated steam to form Silicon(IV) oxide and hydrogen gas in a redox reaction. ✓re

$$Si_{(s)} + 2H_2O_{(g)} \longrightarrow SiO_{2(s)} + 2H_{2(g)}$$ ✓eq

Chlorine being an oxidizing agent, it disproportionates in water to form chloric(I) acid and hydrochloric acid in a redox reaction.

$Cl_{2(g)} + H_2O_{(l)} \longrightarrow HOCl_{(aq)} + HCl_{(aq)}$

Phosphours and sulphur do not react with water at room temperature. This is because their chemical bonds are too strong to be broken by water and also lack the high electron affinity or oxidizing power needed to strip electrons from water molecules.

Reaction with dilute acids,

Sodium being reducing, it reacts with cold, dilute hydrochloric acid to form sodium chloride and hydrogen gas in a redox reaction.

$2Na_{(s)} + 2HCl_{(aq)} \longrightarrow 2NaCl_{(s)} + H_{2(g)}$

Magnesium, being a reducing agent, it reacts with cold dilute hydrochloric acid to form magnesium chloride and hydrogen gas in a redox reaction.

$Mg_{(s)} + 2HCl_{(aq)} \longrightarrow MgCl_{2(aq)} + H_{2(g)}$

Aluminium being highly reducing, it reacts with cold dilute hydrochloric acid in a redox reaction to form Aluminium chloride and hydrogen gas.

$Al_{(s)} + 3HCl_{(aq)} \longrightarrow AlCl_{3(aq)} + \frac{3}{2}H_{2(g)}$.

Phosphorous, Silicon, Sulphur, chlorine do not react with dilute hydrochloric acid. This is because of their chemical bonds too strong to be broken by dilute hydrochloric acid.

With dilute Sulphuric acid.

Sodium being a reducing agent, it react with cold dilute sulphuric acid to form sodium sulphate and hydrogen gas in a redox reaction.

$$2Na_{(s)} + H_2SO_{4(aq)} \longrightarrow Na_2SO_{4(aq)} + H_{2(g)}$$

Magnesium being a reducing agent, it reacts with cold dilute sulphuric acid to form magnesium sulphate and hydrogen gas in a redox reaction

$$Mg_{(s)} + H_2SO_{4(aq)} \longrightarrow MgSO_{4(aq)} + H_{2(g)}$$

Aluminium, sulphur, Silicon, chlorine do not react with dilute sulphuric acid Reason.

With dilute nitric acid.

Magnesium being a reducing agent, it reacts slowly with cold dilute nitric acid to form magnesium nitrate and hydrogen gas in a redox reaction

$$Mg_{(s)} + HNO_{3(aq)} \longrightarrow Mg(NO_3)_{2(aq)} + H_{2(g)}$$

~~Magnes~~

Reaction with alkalis

Sodium and magnesium do not react with alkalis at any condition. This is because of their basic properties being a reducing agent

Aluminium, reacts with hot concentrated sodium hydroxide solution in a redox reaction to form aluminate ions and hydrogen gas

$2Al_{(s)} + 2\bar{O}H_{(aq)} + 6H_2O_{(l)} \longrightarrow 2[Al(OH)_4]^-_{(aq)} + 3H_{2(g)}$

Silicon being a reducing agent, it reacts with both cold dilute sodium hydroxide solution and hot concentrated sodium hydroxide solution to form silicate ions and hydrogen gas in a redox reaction.

$Si_{(s)} + 2\bar{O}H_{(aq)} + H_2O_{(l)} \longrightarrow SiO_{3(aq)}^{2-} + 2H_{2(g)}$

Phosphorous being reducing, reacts with hot concentrated sodium hydroxide solution to form phosphine and phosphinate ions in a redox reaction.

$P_{4(s)} + 3\bar{O}H_{(aq)} + 3H_2O_{(l)} \longrightarrow 3H_2PO^-_{2(aq)} + PH_{3(g)}$

Chlorine being oxidizing, it disproportionates in cold dilute sodium hydroxide solution in a redox reaction to form chloride ions and chlorate(I) ions

~~$Cl_{2(g)} + \bar{O}H_{(aq)} \longrightarrow H(l_{(aq)} + \bar{O}Cl_{(aq)} + H_2O$~~

$Cl_{2(g)} + 2\bar{O}H_{(aq)} \longrightarrow Cl^-_{(aq)} + \bar{O}Cl_{(aq)} + H_2O_{(l)}$

Chlorine being oxidizing, it disproportionates in hot concentrated sodium hydroxide solution in a redox reaction to form chloride ions and chlorate(V) ions

$3Cl_{2(g)} + 6\bar{O}H_{(aq)} \longrightarrow 5Cl^-_{(aq)} + ClO^-_{3(aq)} + 3H_2O_{(l)}$

Sulphur

HEALTH DANGERS:

Na, Al, Mg, Si, P, S, and Cl_2

chlorine is highly toxic thus can damage the eye, skin and lungs when too much exposed, Mitigated wearing protective gears when handling chlorine such as protective clothings, face shield and among others.

Sulphur dust can irritate the eyes and the respiratory tract. Mitigated by wearing protective clothings

Combustion of sulphur produce toxic gases, such as sulphur dioxide which harms the respiratory system thus leading to death. This can be mitigated by use of converters such that there is no escape of gas.

Accept other alternatives

Phosphorous being highly toxic and can cause severe burns when exposed and damages other organs. Mitigated by avoiding too much exposure when using it ie. wear protective gear such as protective clothings.

Aluminium fine dust can irritate the lungs, due to prolonged exposure to high levels thus affecting the nervous system. Mitigated by wearing protective gears such as protective face masks, lead clothing when using the element.

Sodium on reaction with water cause severe burns ingestion can cause poisoning, Mitigated by wearing protective gears such as glover, goggles face shields and many others

Magnesium dust can irritate the respiratory system, burning magnesium produce intense light that can damage the eye. Mitigated wearing Protective gears. such that there is reduction in direct exposure to the metal itself.

Accept any one

TOTAL SCORES: 25

ACCEPT 20 SCORES

ITEM 4

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Trend on atomic radius

Atomic radius increases from flourine to Iodine.

From flourine to Iodine, there is an increase in both nuclear charge and screening effect. This is because from one element to another, there is addition of a fully filled energy level which increases on the screening effect more than the nuclear charge ✓ Te

Trend on boiling point.

Boiling point increases from flourine to Iodine This is because atomic radius increases, molecular size and molecular mass hence the magnitude radius also increases thus increase in the magnitude of van der waal's forces of attraction thus increase in boiling point. This is because an increase in molecular mass increases on the van der waal's forces of attraction thus making the bond strong hence requiring a lot of energy to break thus high boiling point. Te.

Reaction with water.

Reject without reason for not reacting

Iodine is soluble in water. This is because Iodine is non-polar yet water is polar. Therefore, the Intermolecular force of attraction between the Individual molecules of water and Individual molecules of Iodine are on average stronger than the Intermolecular forces of attraction between molecules of water and molecules of Iodine hence when the two molecules combine, replusion occurs

Flourine being an oxidising agent, it reacts vigprously with water in a redox reaction forming oxygen and hydrofluoric acid.

Re eq

$$2F_{2(g)} + 2H_2O_{(l)} \longrightarrow 4HF_{(aq)} + O_{2(g)}$$

Bromine being an oxidising agent, it reacts moderately with water to form bromic(I) acid and hydrobromic acid in a redox reaction.

Re eq

$$Br_{2(l)} + H_2O_{(l)} \longrightarrow HOBr_{(aq)} + HBr_{(aq)}$$

Chlorine being an oxidizing agent, it reacts moderately with water forming hydrochloric acid and chloric(I) acid. in a redox reaction.

Re eq

$$Cl_{2(g)} + H_2O_{(l)} \longrightarrow HOCl_{(aq)} + HCl_{(aq)}$$

Reaction with alkalis

Reaction with hot concentrated sodium hydroxide solution.

Flourine, an oxidizing agent reacts with hot concentrated sodium hydroxide solution to form water, oxygen and fluoride ions in a redox reaction.

Re eq

$$2F_{2(g)} + 4OH^-_{(aq)} \longrightarrow 2H_2O_{(l)} + 4F^-_{(aq)} + O_{2(g)}.$$

Chlorine is highly oxidizing, it reacts with hot concentrated sodium hydroxide solution in a disporportionation reaction forming chloride ions, chlorate (V) ions and water.

$$3Cl_{2(g)} + 6\bar{O}H_{(aq)} \longrightarrow 5Cl^-_{(aq)} + ClO^-_{3(aq)} + 3H_2O_{(l)}$$

Bromine, an oxidising agent reacts with hot concentrated sodium hydroxide solution in a disporportionation reaction to form bromide ions, bromate (V) ions and water.

$$3Br_{2(l)} + 6\bar{O}H_{(aq)} \longrightarrow 5Br^-_{(aq)} + BrO^-_{3(aq)} + 3H_2O_{(l)}$$

Iodine being an oxidising agent, reacts with hot concentrated sodium hydroxide solution in a disporportionation reaction to form Iodide ions, water and Iodate (V) ions

$$3I_{2(s)} + 6\bar{O}H_{(aq)} \longrightarrow 5I^-_{(aq)} + IO^-_{3(aq)} + 3H_2O_{(l)}$$

Reaction with cold dilute sodium hydroxide solution.

Iodine, an oxidising agent reacts with cold dilute sodium hydroxide solution to form Iodide ions, Iodate (I) ions and water in a disporportionation reaction.

$$I_{2(s)} + 2\bar{O}H_{(aq)} \longrightarrow I^-_{(aq)} + IO^-_{(aq)} + H_2O_{(l)}$$

Flourine being an oxidizing agent, reacts with cold dilute sodium hydroxide solution to form Flouride ions, water and oxygen difflouride in a disporportionation reaction.

$$2F_{2(g)} + 2\bar{O}H_{(aq)} \longrightarrow 2F^-_{(aq)} + OF_{2(g)} + H_2O_{(l)}$$

Chlorine being an oxidising agent reacts with cold dilute sodium hydroxide solution to form chloride ions, chlorate (I) ions and water in a disporportionation reaction.

$$Cl_{2(g)} + 2\bar{O}H_{(aq)} \longrightarrow Cl^-_{(aq)} + ClO^-_{(aq)} + H_2O_{(l)}$$

Bromine being an oxidising agent, it disportionates

cold dilute sodium hydroxide solution forming bromide ions, bromate(I) ions and water ✓ee.

$Br_{2(l)} + 2OH^-_{(aq)} \longrightarrow Br^-_{(aq)} + BrO^-_{(aq)} + H_2O_{(l)}$ ✓eq

HEALTH IMPLICATIONS

Flourine

Being highly reactive with water, causes severe burns thus leading to severe injuries. Mitigated by use of protective gears such as slaph-suit, face shield mask. Ventillation ✓Em.

Chlorine

Being a toxic gas, irritates the eye and respiratory system which to much exposed thus damaging the lungs. Mitigated by wearing protective gears, such as face shields, goggles, gloves and ensuring proper ventilation of the gas ✓Em.

Iodine.

Prolonged contact may lead to accidental burn on the skin, irritation. Mitigated by use of protech gears such as goggles, face shield ✓Em.

Accepting one element

Bromine.

Being highly toxic and corrosive, irritate the eyes, skin and may damage the respiratory tract. When exposed to it. Mitigated by keeping it in properly labelled tighted closed containers. Wearing Protective gears for example face shield, protective clothings ✓Em.

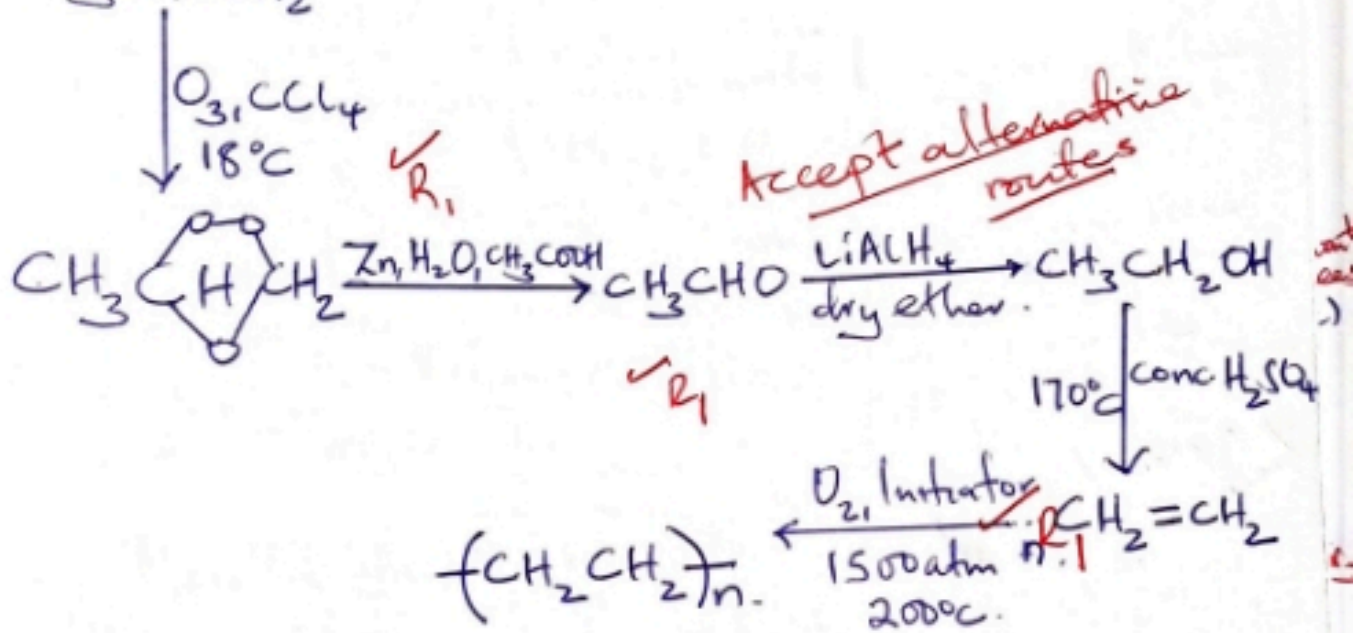
Accept other alternatives

TOTAL SCORES: 25

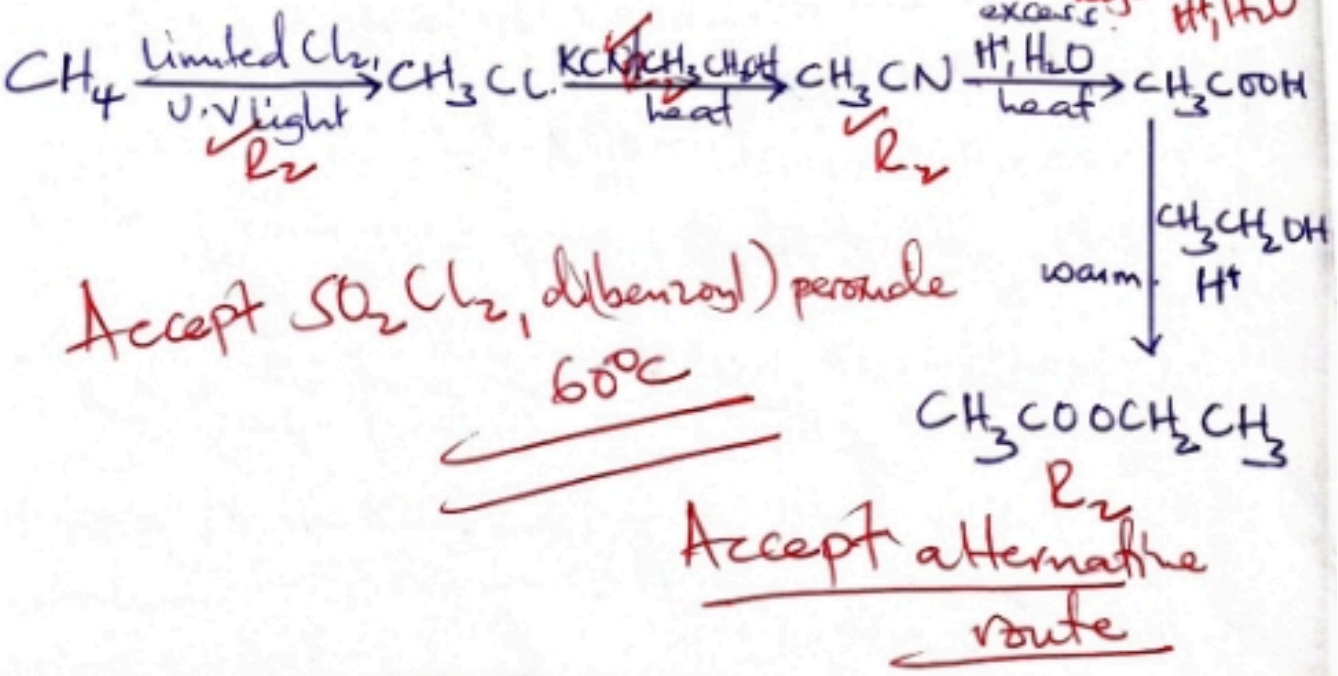
ACCEPT 20 SCORES

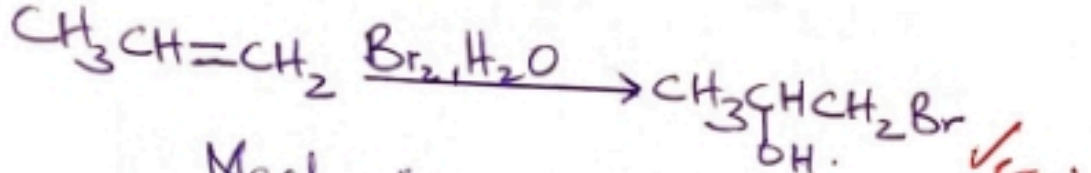
ITEM 5

$-(CH_2-CH_2)_n-$ from $CH_3CH=CH_2$



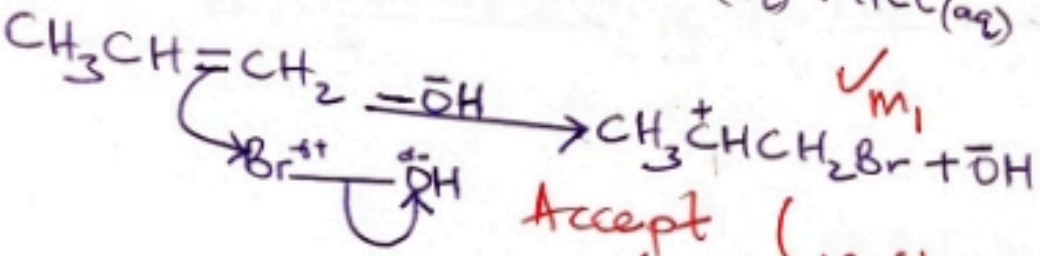
CH_4 to $CH_3COOCH_2CH_3$



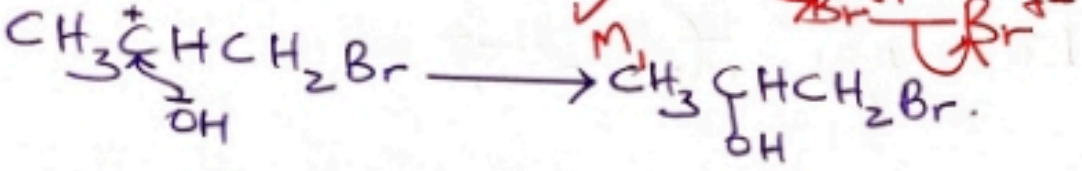


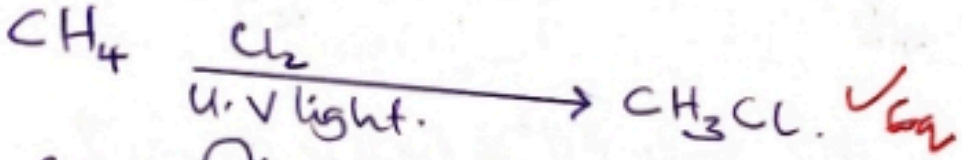
Mechanism:

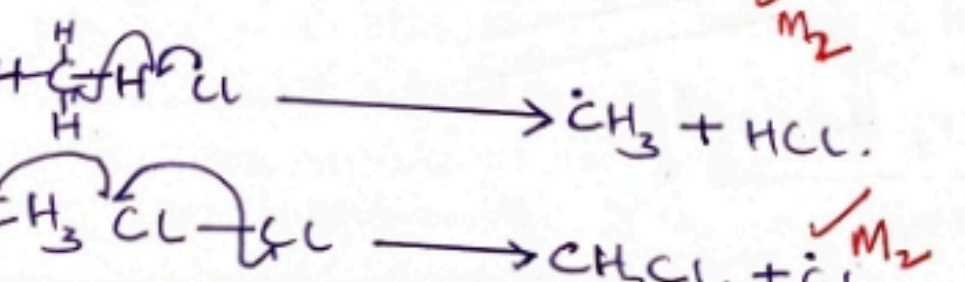




Accept







Suitability

Ethyl acetate being an ester, it acts as an excellent solvent because it dissolves nail polish quickly thus removing polishes.

Polyethene is highly water proof, making it suitable for containers and packages.

Polyethene being strong and durable, it makes it long lasting.

Accept other alternatives

Health implications

Prolonged exposure to the product, ethyl acetate may lead to health diseases which may affect the liver, kidney and the central nervous system leading to death. Mitigated by avoiding prolonged exposure to the product.

The product, ethyl acetate being high flammable, can cause fires leading to skin burn of the workers thus leading to severe injuries. Mitigated by using the in convection using well-ventilated areas to prevent inhalation of vapours or. Installing fire extinguishers where the product is kept.

Combustion of polyethene in air, releases toxic gases such as carbondioxide which is a green house gas causing global warming hence increase in earther temperature. Mitigated by

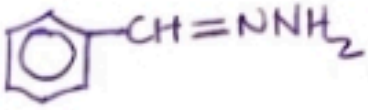
Incomplete combustion of polyethene produces carbon monoxide which is a poisonous gas that reduces the blood ability to carry enough oxygen causing headache leading to death. Mitigated by recycling Polyethene into new products such as plastic containers and many others OR. Avoid Open burning of the product such that their is no release of carbon monoxide OR. ~~and~~ carbondioxide

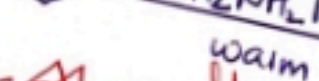
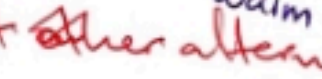
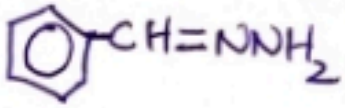
TOTAL SCORES : 25

ACCEPT 20 SCORES

ITEM 6

Ethane to benzaldehyde hydrazone.

CH_3CH_3 to 

CH_3CH_3 — ✓R1 Limited Cl_2, UV light → CH_3CH_2Cl — $NaOH_{(aq)}$, heat, ✓R1 → CH_3CH_2OH — conc. H_2SO_4, 180°C → $H_2C=CH_2$ — Br_2/CCl_4 → $BrCH_2CH_2Br$ — ✓R1 heat, excess NaOH(aq) → CH_3CH_2OH — ✓R1 $CH_3Br/AlBr_3$ → $HC\equiv CH$ — ✓R1 Accept 50°C $CH_3Br/AlBr_3$ →  — $MnO_{2(s)}$, H^+, heat →  — NH_2NH_2/H^+, warm → 

Accept SO_2Cl_2; dibenzoyl peroxide. 60°C, 50, 40, 70°C, 80°C

Reject excess Cl_2

Reject without excess

Reject NaOH(l).

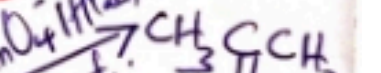
Reject $MnO_4^-/H^+_{(aq)}$

Accept other alternatives

Accept NH_2NH_2

warm.

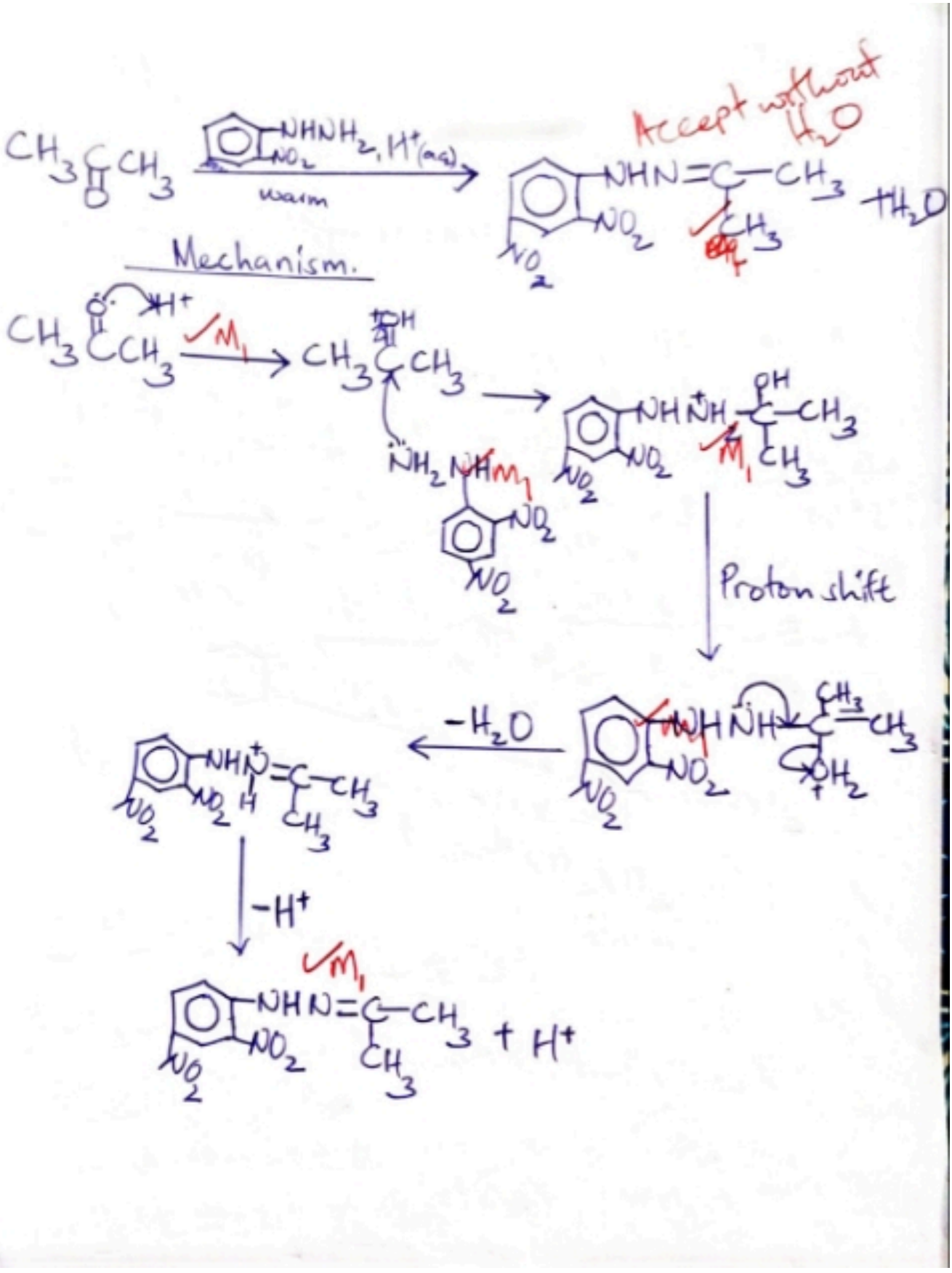
$CH_3CH_2CH_2OH$ to 

$CH_3CH_2CH_2OH$ — 180°C, conc. H_2SO_4, ✓R2 → $CH_3HC=CH_2$ — H^+, H_2O, heat, ✓R2 →  — ✓R2 $MnO_4^-/H^+_{(aq)}$, heat → 

Accept another alternatives

Reject heat.

Accept $MnO_{2(s)}/H^+$. $Cr_2O_7^{2-}/H^+$.


Accept without H2O
warm
Mechanism.
Proton shift
-H2O
-H+

Reject heat.

$$CH_3CH_2CH_2OH \xrightarrow[180^\circ C]{\text{conc. } H_2SO_4} CH_3CH=CH_2 + H_2O$$

✓M_2

Mechanism.

$$H_2SO_4 \rightleftharpoons H^+_{(aq)} + HSO_4^-{}_{(aq)}$$ ✓M_1

OSO_3H

$$CH_3CH_2CH_2\ddot{O}H \longrightarrow CH_3CH_2CH_2\overset{+}{O}H_2$$ M_2

M_2 H^+

Reject if stability is not shown

$-H_2O$

hydride shift.

$$CH_3\overset{+}{C}HCH_3 \rightleftharpoons CH_3CH_2\overset{+}{C}H_2$$ ✓M_2

$+ H_2O$.

$$CH_3\overset{+}{C}H-CH_2 \longrightarrow CH_3CH=CH_2 + H_2SO_4.$$

HSO_4^- ✓M_2

Suitability of Propan-1-ol.

Propan-1-ol is a good organic solvent, so it can be used as the mitigation measure since it dissolves quickly.

Propan-1-ol being completely miscible with water and other organic solvents, it can be used in many chemical processing.

Accept another alternatives

Health Implications of ~~Deans~~ Propan-1-ol.

Propan-1-ol being highly flammable, it can cause irritation on eyes, skin and other respiratory problems when exposure or on use. This can be mitigated by wearing Protective gears such as face shield, face masks, Protective clothings and many others.

Prolonged exposure to propan-1-ol may affect the central nervous system leading to destruction. Mitigated by wearing protective gears when exposed to it.

Propan-1-ol being highly flammable, it can cause fire accidents leading to severe injuries and death. Mitigated by Installing fire extinguishers where the Product is being used.

TOTAL SCORE: 25

H:O:L:D:E:R: N:E:S:S

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